



# Smart 3D LiDAR

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For Security Applications

**SENSTAR™**

# Smart 3D LiDAR is a security product suitable for a wide range of industries

Examples of industries and applications the Smart 3D LiDAR is used in

## Aviation



## Energy and Power



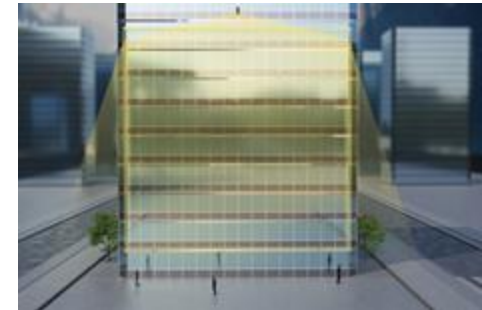
## Buildings/Residential



## Military and Government



## Other use cases



### Airports / Hangars

- Automated exits
- Fence, gates
- Safeguarding the air/land border
- Asset protection

### Energy / network operators

- Power plants, solar parks, transformer stations
- Fence; facade; roof

### Overall building protection

- Highly valuable private properties
- Driveways, entrances, and facades
- Indoor applications

### Military applications

- High security buildings / areas
- Drones / UAVs
- Critical infrastructure

### Security system for endless application areas

# High dependencies on external factors cause high costs and risks

Current challenges in the area of perimeter security

## Established technologies



Camera



Motion sensor



2D scanner



Thermal camera

## Main problems



**Weather conditions** cause false alarms



**Small animals** trigger unwanted alarms



**Lighting conditions** disable common security cameras temporarily



**High number of false alarms** cause a security threat with **high cost for guard interventions**



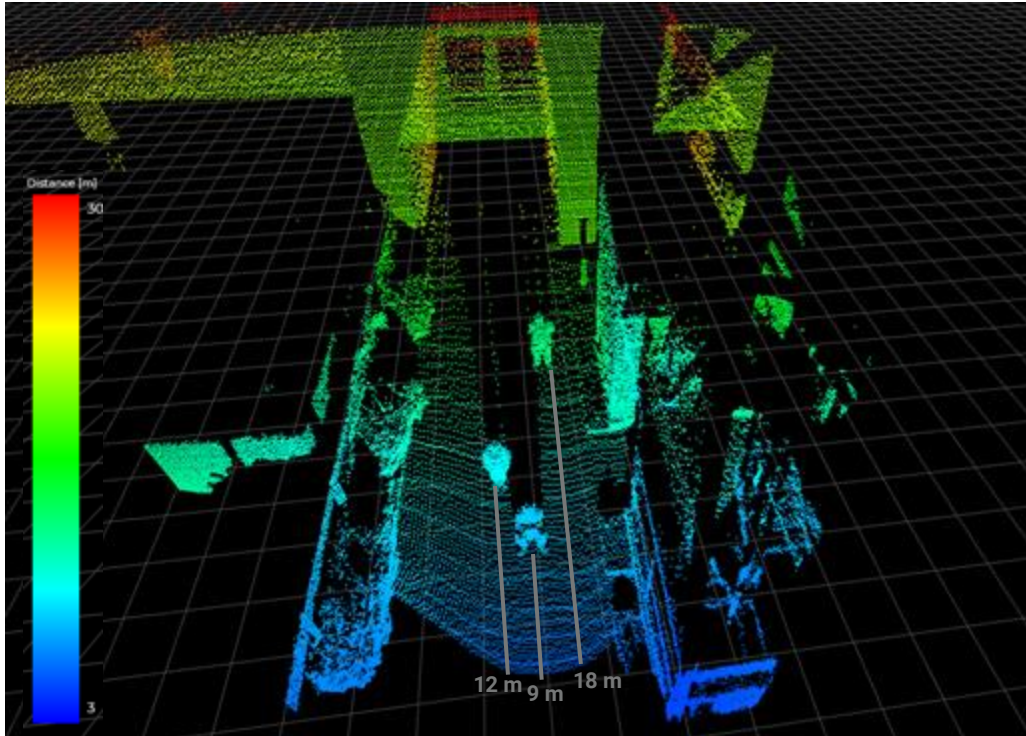
**Privacy Protection** while giving visible feedback is not possible



# Introduction to LiDAR: 3D point cloud view

3D data allows a new perspective

True position in 3D space can be measured directly



Determination of object size independently of the distance



# 3D LiDAR has significant advantages over established technologies

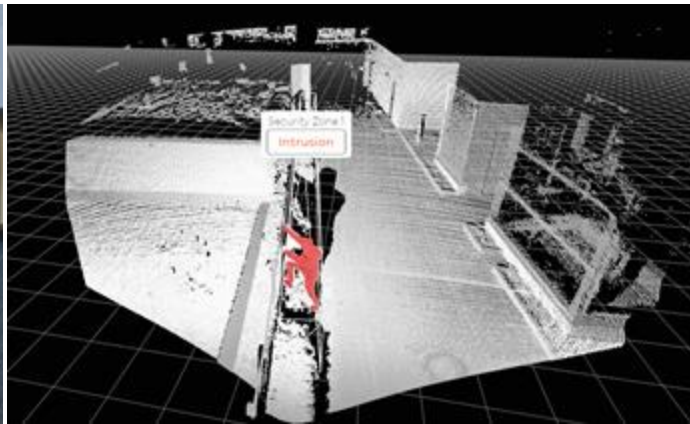
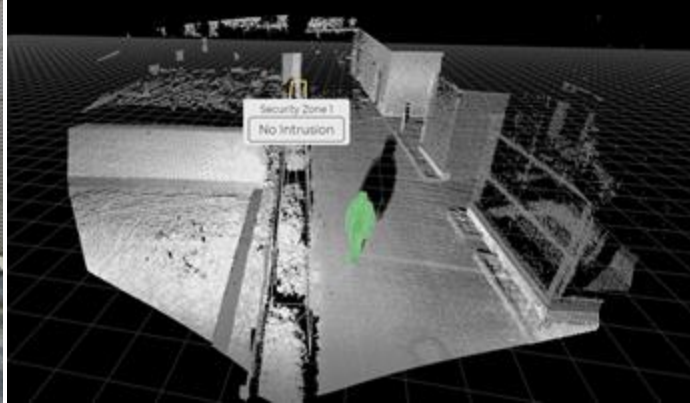
Camera recording vs. LiDAR point cloud

## Camera Recording:



Camera recordings

## LiDAR Point Cloud:



Smart 3D LiDAR LiDAR point clouds with security zones and intrusion detection

## Advantages of 3D LiDAR technology

- ✓ Detection in harsh environments  
IP67, robust against strong sunlight, rain, snow, fog, etc.; day & night, in-& outdoor
- ✓ Privacy protecting with anonymized 3D data
- ✓ Easy installation with PoE;  
Camera-like handling with high-resolution, visual feedback
- ✓ Low maintenance;  
No rotating components, no friction or wearing out, camera-like robustness

# Smart 3D LiDAR Specifications and Accessories

Portfolio offers a wide range of accessories for various application scenarios

## Smart 3D LiDAR



### TECHNICAL DATA

|                           |                                                |
|---------------------------|------------------------------------------------|
| Field of view (H x V)*    | 90° x 50°                                      |
| Range**                   | 85 m                                           |
| Frame rate                | 1 – 50 Hz depending on configured scan pattern |
| Ingress protection        | IP67                                           |
| Dimensions (H x W x D)*** | ca. 75 mm x 111 mm x 83 mm                     |
| Weight***                 | ca. 535 g                                      |
| Voltage input             | Power over Ethernet (PoE); IEEE 802.3at Type 1 |
| Power consumption         | Typ. 10 W; max. 13 W                           |

\* Configurable via web user interface

\*\* Human Target (size 150x50cm; 25% reflectivity); Environmental conditions may influence range performance

\*\*\* Without cables or antenna attached

### ACCESSORIES

Weather Protection Roof



Dual sensor mount



Cabling (PoE)



WiFi antenna (optional)





# On-device software for real-time intrusion detection

Advantages of Smart 3D LiDAR's intuitive software running on-device

## Precise 3D zone placement

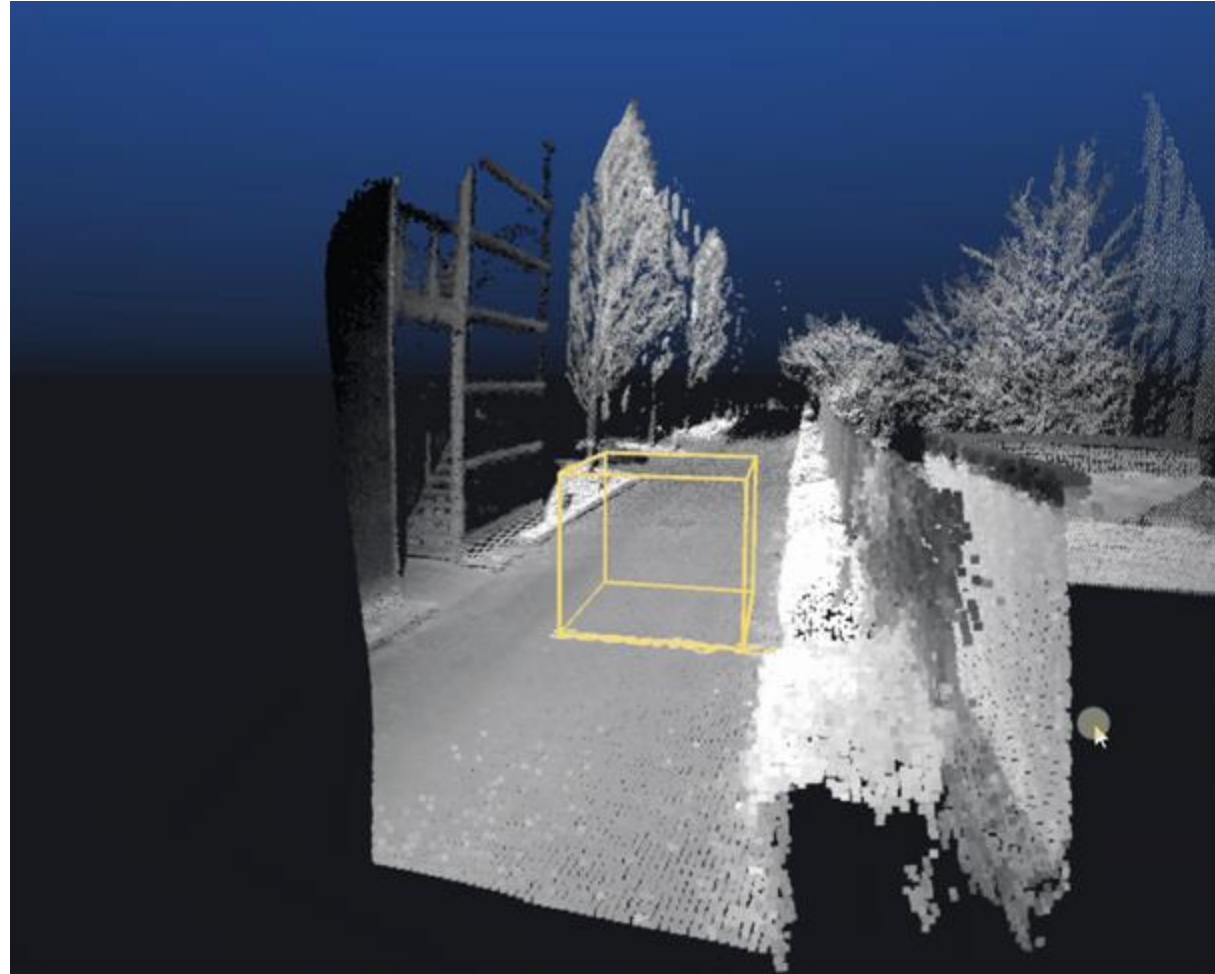
- Drag & drop security zones
- Direction sensitive
- Multiple zones, e.g. (pre-)alarm zone, early detection of suspicious behavior

## Rule-based alarm generation

- Alarms based on true object-size to filter out small animals
- Exclude sources of false alarms like vegetation
- Customizable alarm flows

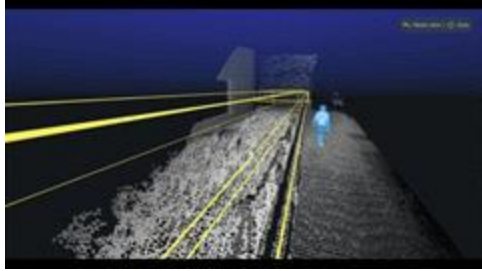
## Standard interfaces

- TCP/IP & MQTT => VMS
- TCP/IP & MQTT => Web I/Os
- ONVIF compatibility



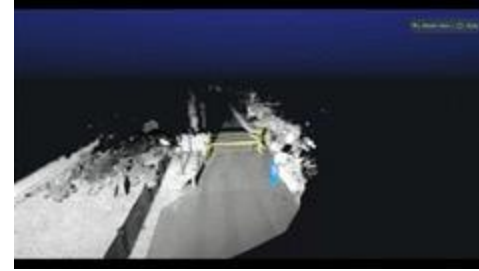
# Offering a cutting-edge solution for precise and reliable security protection.

## Security Applications



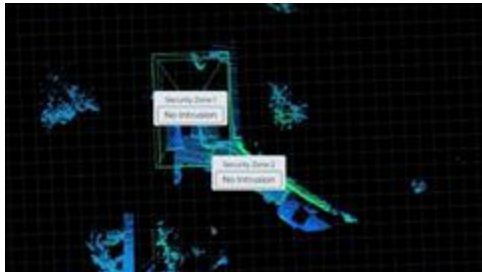
### Fence Monitoring

LiDAR detects intruders, accurately measuring their size and movements. It identifies suspicious behavior in pre-alarm zones and provides immediate alerts if the fence is climbed over or bypassed.



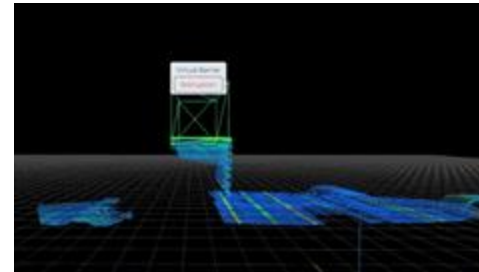
### Gate Monitoring

LiDAR systems monitor barriers and entrance gates, ensuring access is restricted to designated entry points while triggering alarms for any attempts to breach unauthorized areas.



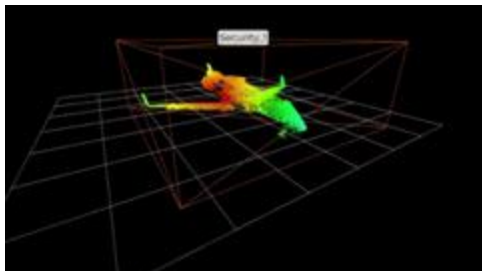
### Intruder Detection

LiDAR is able to detect unauthorized entry and triggering alarms. This technology offers precise monitoring while offering privacy protection at the same time.



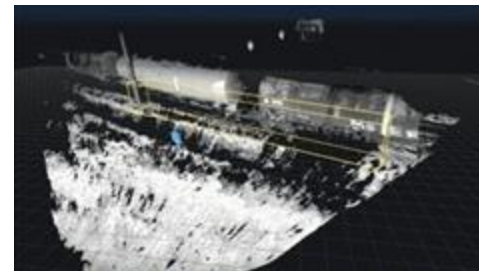
### Traffic Monitoring

LiDAR creates virtual barriers to detect e.g. vehicle heights allowing for smart redirection and enabling precise tracking and management of vehicle movement.



### Asset Protection

LiDAR secures valuable assets by creating 3D zones that allow for precise detection of unauthorized access or movement, even in the absence of physical barriers.



### Railway Surveillance

LiDAR monitors railway crossings for safety and ensure comprehensive oversight of railway areas to maintain security and operational efficiency for safeguarding critical railway infrastructure.



# Maximum security, as discreetly as possible

VIPs, residential and building security

Private property



## Challenges



Property location and design require individual solutions to maintain integrity



No guarantee of privacy through video cameras



Low tolerance for false alarms

## Advantages



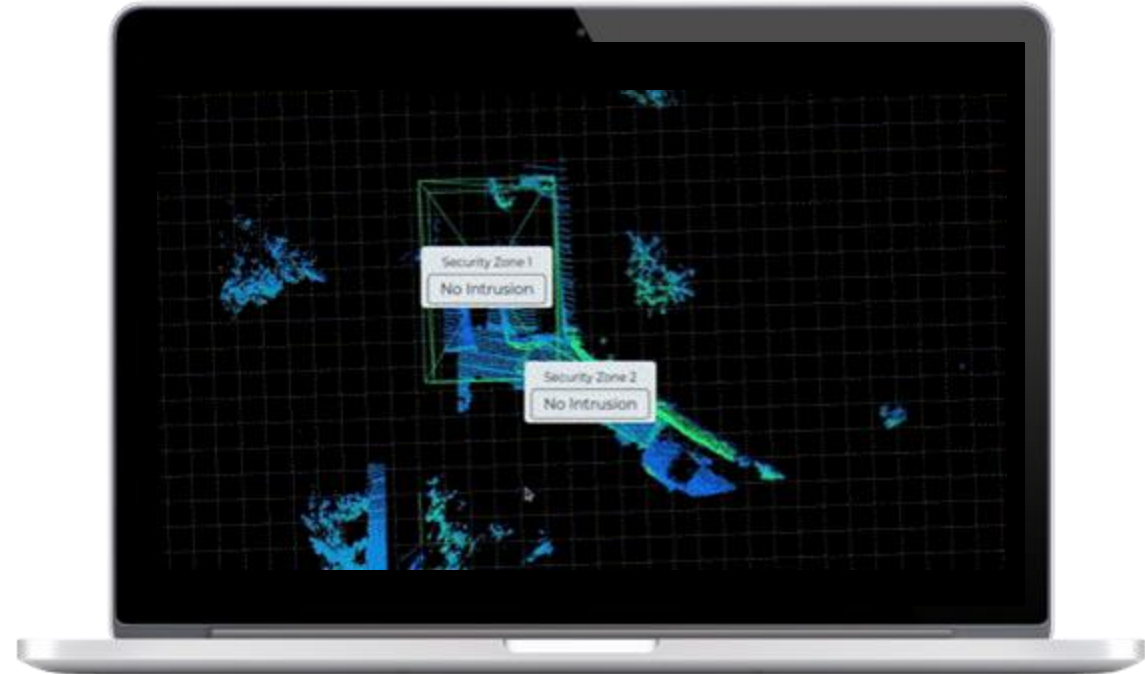
**Compact and unobtrusive design** with minimal cabling effort



Proactive protection through **pre-alarms in the event of suspicious behavior**



Real-time alarm generation enabling **fast response times**



# Automatic locking system with robust roller shutters prevents blasting

Securing bank foyers

## Challenges



**Increasing number** of self-service ATMs blown up with high losses



Solutions such as **ink or adhesive cartridges** are **very expensive**



**High personnel costs** for manual checking of self-service areas

## Advantages



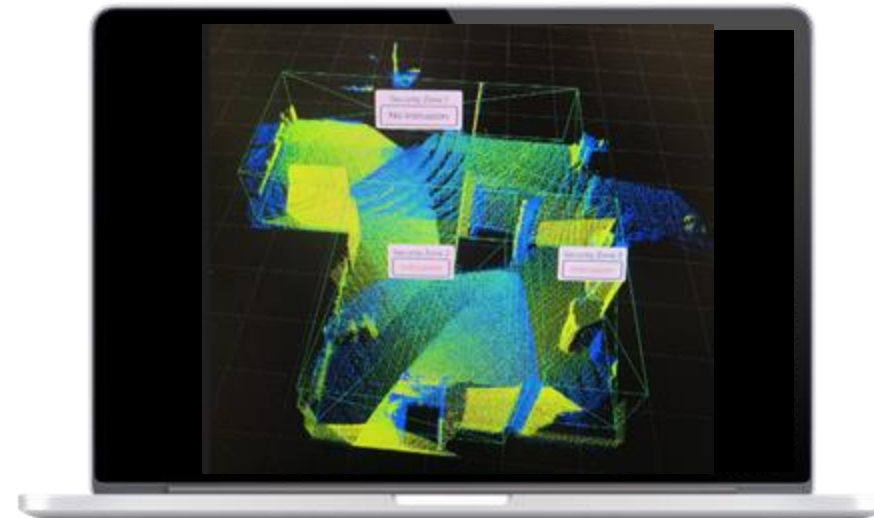
**Reliable real-time detection** of the indoor area and object size



**No manipulation** due to built-in detection of tampering and malfunction



**Process automation** and increased safety thanks to roller shutter door solution



# Ensuring reliable airport security

Securing taxiways, aircraft and access gates

Aviation



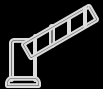
## Challenges



Aviation Security Act: **§8 and §9** require comprehensive security systems and measures



**Large safety areas** with various road users in wind and weather



**Manual security checks** to prevent unauthorized access, e.g. at freight and service gates

## Advantages



Reliable differentiation of objects based on true object-size



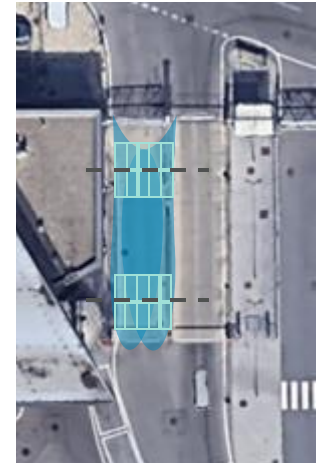
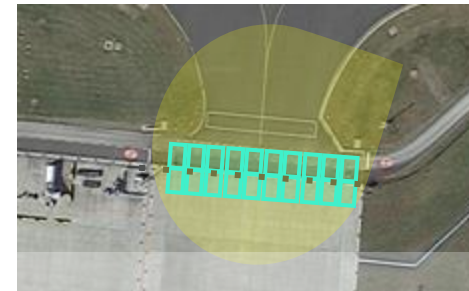
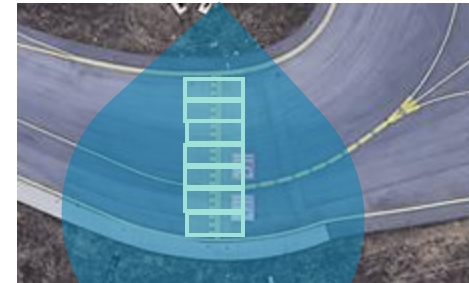
Precise localization and direction of movement



Durable, robust security solution with minimal maintenance requirements



Automated access control reduces risk and costs



# Protection of military facilities

Volumetric protection of objects, hangars and perimeters

Aviation



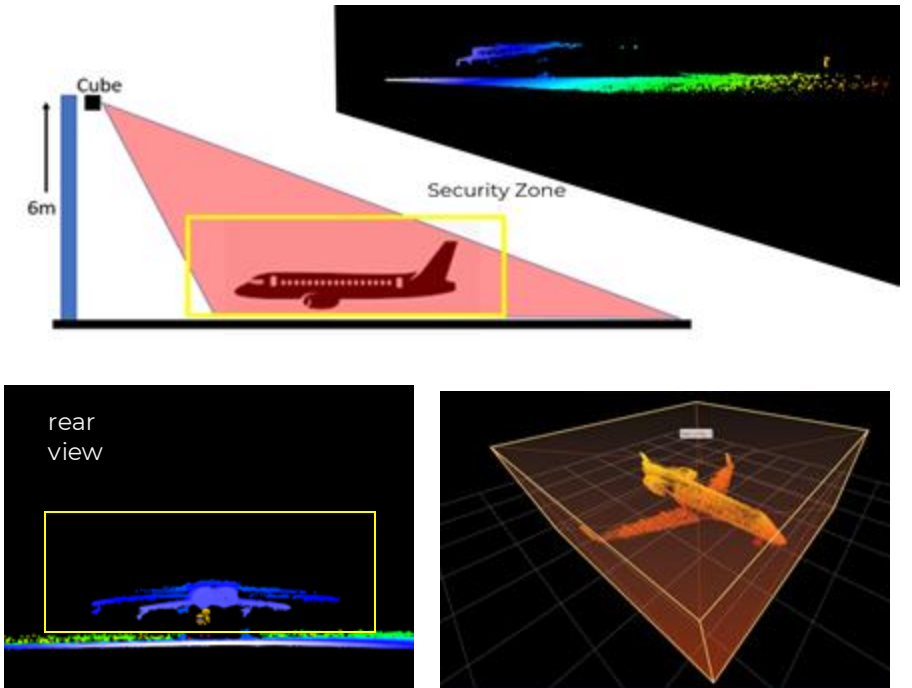
## Challenges



- Reliable detection in difficult environmental conditions (rain, snow, fog, etc.)
- Turbulent scene causes false alarms due to a wide variety of objects (e.g. airplanes, helicopters, tankers, boarding ladders, etc.)

## Advantages

- Automatic teach-in of static objects within the safety zone
- Weather-independent, day & night operation
- Simple installation outdoors: PoE eliminates the need for additional network cabling
- On-device data processing with direct alerting of the customer system (e.g. VMS)





# Physical protection of critical infrastructures

Substations, power plants, high-voltage grids and solar parks

Utilities



## Challenges



**High false alarm rate and verification effort** due to external factors (e.g. weather, vehicles, animals)



**Data protection problems** with systems in urban areas



**High damage** in the event of theft / production downtime (power failure)

## Advantages



**Fast response to real threats** thanks to precise size detection, even in difficult environments



**Seamless integration** into existing security systems and existing infrastructure (e.g. masts)



**Anonymized 3D data** does not allow any conclusions to be drawn about recorded persons

# Reliable high-tech solution with easy & fast transition

## Smart 3D LiDAR - The Smart 3D Security LiDAR

### Why choose Smart 3D LiDAR



#### **Embedded software running on-device**

no additional fees; no external computer required



#### **Low installation cost; camera-like handling**

no experts required for mounting and configuration



#### **Significant reduction in false alarms**

less guard interventions and savings on personnel cost



#### **Standard interfaces and compatibility with VMS**

easy transition and integration; drop-in replacement for (thermal) cameras



#### **Ongoing training and support**

Continuous software updates and support for partners in various customer use cases

# Thank You.

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